ECEN-601: Homework 3

Due on September 26, 2025 at 11:59 pm

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Problem 1

Prove that if a Cauchy sequence in a metric space (not assumed to be complete) has a limit point, then it has a limit. [A point x is a limit point of $\{x_n\}$ if every neighborhood of x contains infinitely many elements of the sequence.]

Solution

In a metric space, a sequence $x_n \in X$ is Cauchy if $\forall \varepsilon > 0$, $\exists N \in \mathbb{Z}$, such that for all $m, n > N$, $d(x_m, x_n) < \varepsilon$.

In order for x to be a limit point of $\{x_n\}$, there must exist some $\varepsilon > 0$, such that for all $m > N$, $d(x_m, x) < \varepsilon$. Since a Cauchy sequence requires a convergence of x_m and x_n, then that same convergence must also apply to x_m and x. Thus, if a Cauchy sequence has a limit point, then it must also have a limit.

Problem 2

Call two metrics d_1 and d_2 on the same set M equivalent if there exist positive constants c_1, c_2 such that $d_1(x, y) \leq c_2 d_2(x, y)$ and $d_2(x, y) \leq c_1 d_1(x, y)$ for all x and y in M. Prove that $x_n \to x$ in the d_1-metric if and only if $x_n \to x$ in the d_2-metric, if d_1 and d_2 are equivalent.

Solution

If $x_n \to x$ in d_1, then $\forall \varepsilon_1 > 0$, $\exists N \in \mathbb{Z}$, such that for all $n > N$, $d_1(x_n, x) < \varepsilon_1$ and if $x_n \to x$ in d_2, then $\forall \varepsilon_2 > 0$, $\exists N \in \mathbb{Z}$, such that for all $n > N$, $d_2(x_n, x) < \varepsilon_2$. By the metric equivalence,

$$d_2(x_n, x) \leq c_1 d_1(x_n, x) \leq c_1 \varepsilon_1$$

If $\varepsilon_1 = \varepsilon_2 / c_1$, then

$$d_2(x_n, x) \leq \varepsilon_2$$

Therefore, $x_n \to x$ in d_2 when $x_n \to x$ in d_1.

Since the reverse argument also holds, $x_n \to x$ in d_1 if $x_n \to x$ in d_2. Thus, $x_n \to x$ in the d_1-metric if and only if $x_n \to x$ in the d_2-metric.

Problem 3

If $f : M \to N$ and $g : N \to P$ are continuous, prove $g \circ f : M \to P$ is continuous. Given an example where $g \circ f$ is continuous but g and f are not.

Solution

By the properties of function composition, the composition of two continuous functions is also continuous.

Example: I will construct a sine wave from two discontinuous functions. Let $f(x) = x \pmod{2\pi}$.

Then, we define some $g(x)$ as,

$$g(x) = \begin{cases} \sin(x) & 0 \leq x \leq 2\pi \\ a & \text{otherwise} \end{cases}$$

which is discontinuous for any $a \neq 0$.

Then, simply $g \circ f = \sin(x)$, which is continuous.

Problem 4

Let a be a fixed real number with $1 < a < 3$. Prove that the mapping $f(x) = (x/2)+(a/2x)$ satisfies the hypotheses of the contractive mapping principle on the domain $(1, \infty)$. What is the fixed point?

Solution

If f satisfies the hypotheses of the contractive mapping principle, then $\exists c \in [0, 1)$, such that $\forall x, y \in M$,

$$d(f(x), f(y)) \leq c \cdot d(x, y)$$

Here, our function is defined as

$$f(x) = \frac{x}{2} + \frac{a}{2x}$$

Then, we have

$$\begin{aligned} d(f(x), f(y)) &= \left| \left(\frac{x}{2} + \frac{a}{2x} \right) - \left(\frac{y}{2} + \frac{a}{2y} \right) \right| \\ &= \left| \frac{x-y}{2} + \frac{a(y-x)}{2xy} \right| \\ &= \left| (x-y) \left(\frac{1}{2} - \frac{a}{2xy} \right) \right| \end{aligned}$$

Here $|(x-y)| = d(x, y)$, so we can write

$$d(f(x), f(y)) = \underbrace{\left| \frac{1}{2} - \frac{a}{2xy} \right|}_{c} \cdot d(x, y) \leq c \cdot d(x, y)$$

The fixed point occurs when $f(x) = x$, thus

$$\begin{aligned} x &= \frac{x}{2} + \frac{a}{2x} \\ 2x^2 &= x^2 + a \\ x^2 &= a \\ x &= \pm\sqrt{a} \end{aligned}$$

We know $1 < a < 3$, so $x = \sqrt{a}$ is the fixed point.

Problem 5

If f satisfies the hypotheses of the contractive mapping principle and x_1 is any point in M, show that $d(x_1, x_0) \leq d(x_1, f(x_1))/(1-r)$ where x_0 is the fixed point. Informally, this says that if $f(x_1)$ is close to x_1, then x_1 is close to the fixed point (but r must not be too close to 1 for this to be a good estimate).

Solution

By the triangle inequality, we have

$$d(x_1, x_0) \leq d(x_1, f(x_1)) + d(f(x_1), x_0)$$

Since x_0 is the fixed point, $f(x_0) = x_0$, thus

$$d(x_1, x_0) \leq d(x_1, f(x_1)) + d(f(x_1), f(x_0))$$

Then, by the contractive mapping principle, we have

$$d(x_1, x_0) \leq d(x_1, f(x_1)) + r \cdot d(x_1, x_0)$$

Rearranging, we get

$$\begin{aligned} d(x_1, x_0) - r \cdot d(x_1, x_0) &\leq d(x_1, f(x_1)) \\ (1-r) \cdot d(x_1, x_0) &\leq d(x_1, f(x_1)) \\ d(x_1, x_0) &\leq \frac{d(x_1, f(x_1))}{1-r} \end{aligned}$$